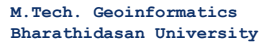


Remote sensing | Satellite Image Processing  
Research Aspirant in GIS & RS



- Academic Project -Coastal Vulnerability Assessment of Kasaragod and Kannur: Using Geospatial Techniques



- Academic Project - *Landslide susceptibility modeling using bivariate analysis and machine learning techniques in Kodaikanal and its environ: a multi-parameter GIS approach (M. Tech – Mini Project)*

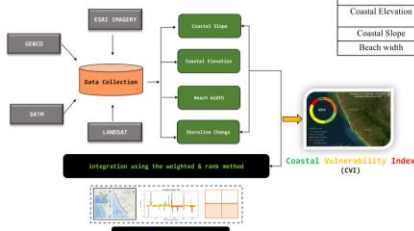
6 Months project/internship

NRSC ISRO

- *Burned Area Extraction Using AI and ML (Deep Learning)* (M. Tech -Major Project)

**M.Sc. Project - Coastal Vulnerability Assessment of Kasaragod and Kannur: Using Geospatial Techniques**

### Adopted Methodology

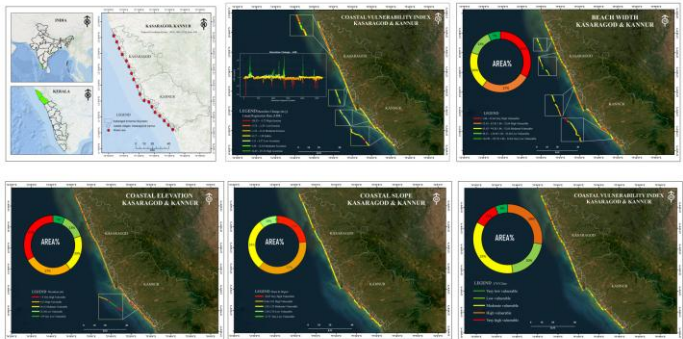


| VARIABLES         | SENSOR/SATELLITE    | TIME PERIOD              | SOURCE              |
|-------------------|---------------------|--------------------------|---------------------|
| Shoreline Change  | TM, ETM+, OLI, TIRS | 1990, 2000<br>2013, 2023 | USGS Earth Explorer |
| Coastal Elevation | SRTM                | *****                    | USGS Earth Explorer |
| Coastal Slope     | ..                  | *****                    | GEBCO               |
| Beach width       | ESRI Imagery        | *****                    | Arc GIS             |

#### Software's And tools used

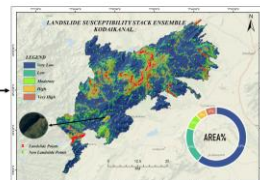
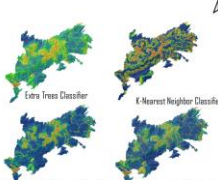
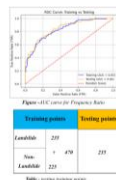
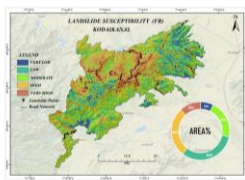
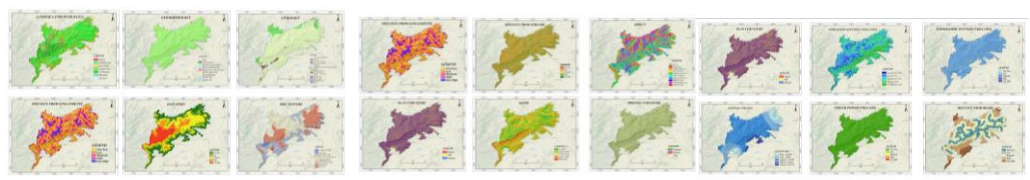
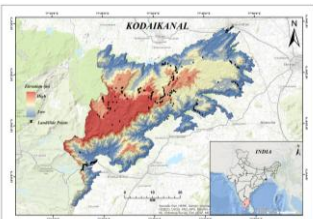
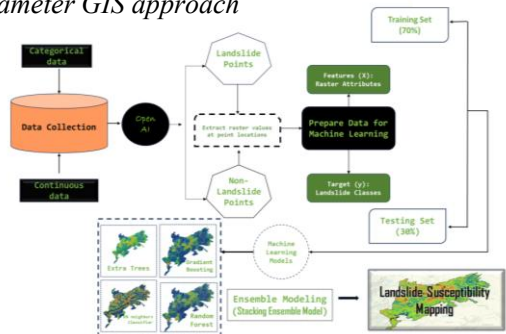
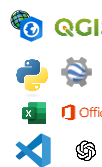
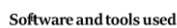
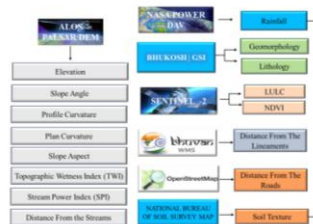


<sup>a</sup>LRR, linear regression rate, which is most commonly used analysis long term shoreline change<sup>20</sup>



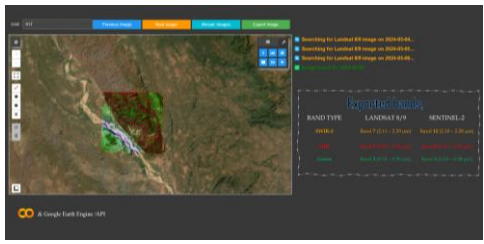
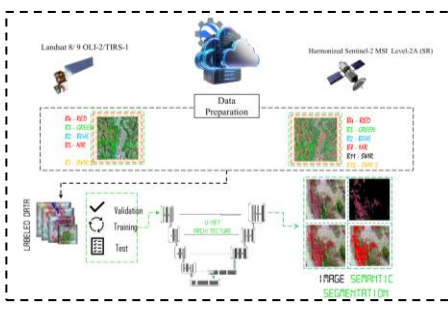
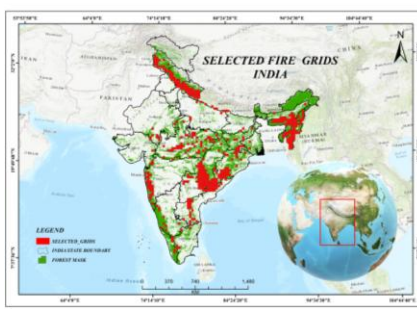
**M. Tech – Mini Project - Landslide susceptibility modeling using bivariate analysis and machine learning techniques in Kodaikanal and its environ: a multi-parameter GIS approach**

### Adopted Methodology



| Susceptibility Zone | No. Of Testing Loadable | Percentage Of Testing Loadable | Avg Percentage |
|---------------------|-------------------------|--------------------------------|----------------|
| Very Low            | 0                       | 0.00                           | 7.44           |
| Low                 | 4                       | 1.50                           | 35.23          |
| Medium              | 35                      | 14.17                          | 53.44          |
| High                | 55                      | 21.65                          | 27.53          |
| Very High           | 149                     | 58.72                          | 13.96          |

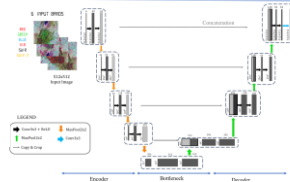
## M. Tech – Major Project - **BURNED AREA EXTRACTION USING AI AND ML (deep learning)**



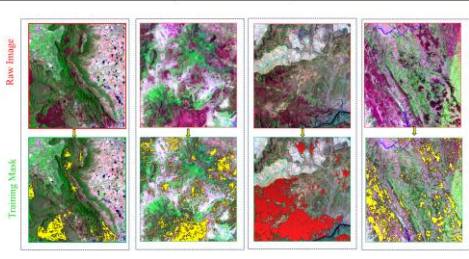
### Software and tools used



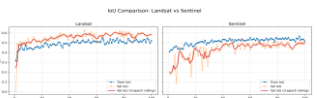
**J NET ARCHITECTURE**



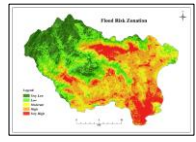
### TRAINING MASKS (LABELED DATA)



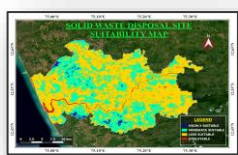
#### MODEL PERFORMANCE EVALUATION: LANDSAT VS. SENTINEL



## *Additional Works*



Book Chapter Co-Author



## Solid waste site suitability in Kasaragod

### *Professional Activities & Achievements*

- Invited as a Resource Person at Kannur University, Department of Geography
- Presented papers at Two International Conferences
- Attended Two Academic Workshops